

HAL CANON

BOOK IX

Interface and Protocol Reference

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AUTHORITY STATEMENT

Subordinate to Books I-IV; semantically aligned with Book X. Book IX defines contracts and does not redesign architecture.

HAL Book IX — Interface and Protocol Reference

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Document Control

Revision History

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1. Purpose, Scope, Authority, and Conformance

Book IX is the canonical contract-level reference for HAL machine interactions. It binds wire behavior to the responsibilities and logical interfaces already approved in Book II and specified in Book IV. It does not create components, move state ownership, or confer authority.

Normative controls

IX-GOV-001 — Authority hierarchy

All contracts **MUST** preserve Books I-IV and X; conflict **MUST** fail closed and be recorded.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-GOV-002 — Contract registration

Every machine interaction admitted to a declared live-effect environment and approved Reality Boundary stage **MUST** use a registered versioned contract.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-GOV-003 — No architectural redesign

A contract **MUST NOT** move responsibility, state ownership, or authority between Book IV components.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

Required practices

Prohibited practices

Required evidence and verification

A conforming release produces schema-validation results, provider and consumer contract-test results, compatibility evidence, security-policy decisions, sample trace linkage, negative-test evidence, and a signed artifact manifest. Violations block release or isolate the affected interaction until corrected.

Examples and anti-patterns

Example: A duplicate command with the same semantic payload and idempotency key returns the original disposition and correlation evidence. Anti-pattern: a timeout causes a new key and a repeated external purchase without reconciliation.

Traceability and review

This chapter implements the applicable Book II interface and trust requirements, Book III design/security/testing controls, Book IV logical-interface handoff, and Book X semantics. Constitutional, architectural, engineering, security/privacy, and semantic review status: PASS. Owner Review items: none.

2. Contract Taxonomy and Catalog

HAL uses commands for requested transitions or effects, queries for non-mutating retrieval, events for completed facts, and streams for ordered sequences. Each registered contract has one provider, an authorized consumer class, a stable identifier, explicit semantics, and a lifecycle state.

Normative controls

This chapter applies controls IX-GOV-001 through IX-GOV-003 as defined once in Chapter 1. The identifiers are cross-references here and are not reissued or independently waivable.

Required practices

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3. Common Message Envelope

The HAL envelope carries typed payloads with identity, contract and schema versions, correlation, causation, time, provenance, classification, integrity, and optional authority and Treaty context. Metadata is part of the security decision, not decorative tracing.

Normative controls

IX-ENV-001 — Common envelope

Every message **MUST** carry message, contract, schema, correlation, causation, producer, time, provenance, classification, and integrity metadata.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-ENV-002 — Freshness

Receivers **MUST** enforce declared issued-at, expiry, and maximum-age constraints.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-ENV-003 — Canonical identifiers

Identifiers **MUST** be opaque, stable, globally unambiguous, and **MUST NOT** embed mutable Authority or Permission claims.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

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4. Identity, Authentication, and Authority Context

Machine identity authenticates an actor; authority context proves the permitted action and constraints. Receivers independently validate current authority. Delegation chains, policy decisions, purpose, target, resource limits, and expirations are explicit and integrity protected.

Normative controls

IX-AUT-001 — Authenticated identity

Every non-public interaction **MUST** authenticate sender and intended recipient.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-AUT-002 — Authority context

Protected interactions **MUST** carry integrity-protected authority, delegation, policy, purpose, and constraint references.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-AUT-003 — No ambient authority

Network location, credentials, data possession, or component execution **MUST NOT** independently confer authority.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-AUT-004 — Decision freshness

Authority decisions **MUST** be current for the proposed action and invalidated by revocation, expiry, or material proposal change.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release

gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-AUT-005 — Denied means no effect

A denial response **MUST** be terminal for the attempted action and **MUST NOT** be converted into success by a transport adapter.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

Required practices

Prohibited practices

Required evidence and verification

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Examples and anti-patterns

Example: A duplicate command with the same semantic payload and idempotency key returns the original disposition and correlation evidence. Anti-pattern: a timeout causes a new key and a repeated external purchase without reconciliation.

Traceability and review

This chapter implements the applicable Book II interface and trust requirements, Book III design/security/testing controls, Book IV logical-interface handoff, and Book X semantics. Constitutional, architectural, engineering, security/privacy, and semantic review status: PASS. Owner Review items: none.

5. Commands, Responses, and Idempotency

Commands express proposed actions, never assumed outcomes. Retriable commands use stable idempotency identities. Results distinguish accepted, committed, denied, pending, indeterminate, and failed. Reality Boundary ambiguity requires reconciliation before retry.

Normative controls

IX-IDM-001 — Idempotency key

Every retrieable command **MUST** require a caller-stable idempotency key scoped to principal, operation, and semantic payload.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-IDM-002 — Duplicate disposition

A duplicate command **MUST** return the recorded disposition and **MUST NOT** repeat the effect.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-IDM-003 — Reality ambiguity

An ambiguous Reality Boundary outcome **MUST** enter reconciliation; automatic replay is prohibited until real-world state is verified.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

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Examples and anti-patterns

Example: A duplicate command with the same semantic payload and idempotency key returns the original disposition and correlation evidence. Anti-pattern: a timeout causes a new key and a repeated external purchase without reconciliation.

Traceability and review

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6. Queries, Pagination, Filtering, and Partial Results

Queries are non-mutating and authority checked. Pagination cursors bind the caller, snapshot, sort, and filter. Partial, redacted, stale, or degraded results are explicit and cannot masquerade as complete authoritative state.

Normative controls

IX-PAG-001 — Pagination

List queries **MUST** use opaque, integrity-protected cursors bound to filter, sort, principal, and snapshot semantics.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-PAG-002 — Filtering

Filterable and sortable fields **MUST** be allowlisted; unrecognized fields **MUST** fail validation.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

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Prohibited practices

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7. Events, Ordering, Delivery, and Replay

Events are immutable completed facts. Contracts state ordering scope and delivery guarantee. At-least-once delivery requires consumer deduplication; replay is labeled, authorized, bounded, and distinguishable from live delivery.

Normative controls

IX-ORD-001 — Ordering scope

Contracts **MUST** declare no ordering, per-key ordering, or total ordering; consumers **MUST NOT** infer stronger order.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-ORD-002 — Event immutability

Events **MUST** describe completed facts and **MUST NOT** be rewritten; correction uses a new linked event.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

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8. Streaming and Flow Control

Streams declare open, data, checkpoint, gap, and close frames; demand, buffering, backpressure, resume, retention, Authority and Permission re-evaluation, and termination behavior are bounded and testable.

Normative controls

IX-STR-001 — Stream resume

Resumable streams **MUST** use opaque cursors with declared retention and gap behavior.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-STR-002 — Stream termination

Streams MUST define normal completion, cancellation, deadline, Authority or Permission loss, Treaty loss, overload, and integrity-failure closure.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

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9. Errors, Denials, Timeouts, Retries, and Cancellation

The stable error model separates validation, authentication, Authority-context failure, Permission denial, conflict, not-found, rate, timeout, dependency, integrity, compatibility, Treaty, and internal failure. Retryability is machine-readable and conservative.

Normative controls

IX-ERR-001 — Structured errors

Errors **MUST** use the HAL error schema with stable code, category, retry disposition, correlation, and detail bounded by classification, privacy, and secret-disclosure controls.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-ERR-002 — No sensitive errors

Error messages **MUST NOT** expose secrets, sensitive payloads, internal credentials, or prohibited inference.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-ERR-003 — Partial result

Partial results **MUST** be explicitly marked with omitted scopes and **MUST NOT** be represented as complete.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

Required practices

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Required evidence and verification

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10. Versioning, Compatibility, and Deprecation

Contract versions follow semantic compatibility rules. Unknown semantics are rejected. Compatibility adapters are named, reviewed components with evidence; they may translate representation but cannot weaken authority or invent missing meaning.

Normative controls

IX-VER-001 — Semantic versioning

Major versions indicate incompatible semantics; minor versions are backward-compatible additions; patches do not change semantics.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-VER-002 — Unknown schema

Unknown or incompatible schemas **MUST** be rejected or handled by an explicitly approved compatibility adapter.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-VER-003 — Consumer tolerance

Consumers **MAY** ignore declared extension fields but **MUST** reject unknown required semantics.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-VER-004 — Deprecation

Deprecation MUST publish replacement, migration evidence, first notice, last supported version, and removal date.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

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11. Security, Privacy, Classification, and Evidence

Contracts minimize data, classify sensitive fields, protect each hop, validate inputs and outputs, prevent replay, constrain telemetry, and generate attributable evidence. Access and disclosure are independently authorized.

Normative controls

IX-SEC-001 — Transport protection

Confidentiality and integrity **MUST** protect every non-public hop with authenticated peer identity.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-SEC-002 — Input validation

Receivers **MUST** validate structure, type, bounds, encoding, canonicalization, classification, current Authority, and the exact Permission decision before use.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-SEC-003 — Replay defense

Protected requests **MUST** include nonce or idempotency identity, time bounds, and integrity evidence sufficient to detect replay.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

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12. External Trust Domains, Treaties, and Constitutional Firewall

Every cross-domain exchange requires authenticated domain identity, active applicable Treaty, permitted purpose/data/capability, and Firewall ingress or egress admission. Revocation, expiry, integrity failure, or drift stops exchange.

Normative controls

IX-TRT-001 — Treaty required

Cross-domain exchange **MUST** cite an active applicable Treaty and **MUST** fail closed if absent, expired, revoked, or materially drifted.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-TRT-002 — Firewall admission

Every external ingress and egress **MUST** pass the Constitutional Firewall; direct bypass is prohibited.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-TRT-003 — Separate approvals

Treaty approval **MUST NOT** substitute for capability-class or action authority approval.

Applicability: All contracts to which this subject applies. **Responsible roles:** provider owner, consumer owner, interface steward, security reviewer. **Enforcement:** schema validation, gateway policy, contract tests, and release gates. **Evidence:** validated artifact, test result, compatibility report, and correlated runtime evidence. **Exception authority:** Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. **Verification:** positive and negative conformance tests.

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Example: A duplicate command with the same semantic payload and idempotency key returns the original disposition and correlation evidence. **Anti-pattern:** a timeout causes a new key and a repeated external purchase without reconciliation.

Traceability and review

This chapter implements the applicable Book II interface and trust requirements, Book III design/security/testing controls, Book IV logical-interface handoff, and Book X semantics. Constitutional, architectural, engineering, security/privacy, and semantic review status: PASS. Owner Review items: none.

13. Observability Contracts and Operational Limits

Contract telemetry provides correlation, outcome, latency, saturation, retry, denial, and integrity signals bounded by classification, privacy, retention, and secret-disclosure controls. Rate, size, concurrency, batch, queue, and stream budgets are declared and enforced without silent truncation.

Normative controls

IX-OBS-001 — Trace correlation

Interactions **MUST** emit correlation, causation, contract version, outcome, latency, and policy-decision references.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-OBS-002 — Evidence integrity

Audit evidence **MUST** be tamper-evident, access-controlled, retention-governed, and attributable.

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IX-OBS-003 — Sensitive telemetry

Telemetry **MUST** use redaction or references rather than unrestricted sensitive payload copies.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

Required practices

Prohibited practices

Required evidence and verification

A conforming release produces schema-validation results, provider and consumer contract-test results, compatibility evidence, security-policy decisions, sample trace linkage, negative-test evidence, and a signed artifact manifest. Violations block release or isolate the affected interaction until corrected.

Examples and anti-patterns

Example: A duplicate command with the same semantic payload and idempotency key returns the original disposition and correlation evidence. Anti-pattern: a timeout causes a new key and a repeated external purchase without reconciliation.

Traceability and review

This chapter implements the applicable Book II interface and trust requirements, Book III design/security/testing controls, Book IV logical-interface handoff, and Book X semantics. Constitutional, architectural, engineering, security/privacy, and semantic review status: PASS. Owner Review items: none.

14. OpenAPI, AsyncAPI, Protocol Buffers, and JSON Schema Profiles

OpenAPI describes request/response bindings; AsyncAPI describes events and channels; Protocol Buffers provide binary service and envelope definitions; JSON Schema is the canonical structural validation profile. Generated bindings remain subordinate to registered semantics.

Normative controls

IX-CNF-001 — Schema validation

Every release **MUST** validate examples and artifacts against the registered schemas.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-CNF-002 — Contract tests

Providers and consumers **MUST** pass positive, denial, malformed, incompatible, replay, timeout, duplicate, and limit tests.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-CNF-003 — Breaking-change gate

Incompatible change **MUST** create a new major contract and coexist through the approved migration window.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

IX-CNF-004 — Evidence manifest

Certification **MUST** bind tested artifacts, hashes, versions, environment, results, and approving roles.

Applicability: All contracts to which this subject applies. Responsible roles: provider owner, consumer owner, interface steward, security reviewer. Enforcement: schema validation, gateway policy, contract tests, and release gates. Evidence: validated artifact, test result, compatibility report, and correlated runtime evidence. Exception authority: Interface Steward with Architecture and Security concurrence; no exception may waive Books I-IV. Verification: positive and negative conformance tests.

Required practices

Prohibited practices

Required evidence and verification

A conforming release produces schema-validation results, provider and consumer contract-test results, compatibility evidence, security-policy decisions, sample trace linkage, negative-test evidence, and a signed artifact manifest. Violations block release or isolate the affected interaction until corrected.

Examples and anti-patterns

Example: A duplicate command with the same semantic payload and idempotency key returns the original disposition and correlation evidence. Anti-pattern: a timeout causes a new key and a repeated external purchase without reconciliation.

Traceability and review

This chapter implements the applicable Book II interface and trust requirements, Book III design/security/testing controls, Book IV logical-interface handoff, and Book X semantics. Constitutional, architectural, engineering, security/privacy, and semantic review status: PASS. Owner Review items: none.

15. Conformance, Compatibility Testing, and Certification Evidence

Provider and consumer conformance covers happy paths and adversarial cases: malformed data, stale authority, denial, duplicates, replay, incompatible versions, timeout, cancellation, overload, Treaty loss, Firewall rejection, redaction, and recovery.

Normative controls

This chapter applies controls IX-CNF-001 through IX-CNF-004 as defined once in Chapter 14. The identifiers are cross-references here and are not reissued or independently waivable.

Required practices

Prohibited practices

Required evidence and verification

A conforming release produces schema-validation results, provider and consumer contract-test results, compatibility evidence, security-policy decisions, sample trace linkage, negative-test evidence, and a signed artifact manifest. Violations block release or isolate the affected interaction until corrected.

Examples and anti-patterns

Example: A duplicate command with the same semantic payload and idempotency key returns the original disposition and correlation evidence. Anti-pattern: a timeout causes a new key and a repeated external purchase without reconciliation.

Traceability and review

This chapter implements the applicable Book II interface and trust requirements, Book III design/security/testing controls, Book IV logical-interface handoff, and Book X semantics. Constitutional, architectural, engineering, security/privacy, and semantic review status: PASS. Owner Review items: none.

16. Contract Catalog and Examples

Each record below formalizes one Book IV logical interface. The common schemas and profiles supply the normative wire fields; provider-specific payload schemas MUST refine, not weaken, them. HTTP routes are canonical bindings for request/response profiles; event channel names are defined by AsyncAPI.

CMP-01 — Constitutional Kernel

All CMP-01 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-02 — Identity and Continuity Service

All CMP-02 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-03 — Authority and Delegation Service

All CMP-03 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-04 — Intent Service

All CMP-04 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-05 — Cognitive Orchestrator

All CMP-05 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-06 — Attention Manager

All CMP-06 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-07 — Judgment Service

All CMP-07 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-08 — Knowledge Service

All CMP-08 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-09 — Memory System

All CMP-09 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV. Restrict Memory Association and Memory Association Restricted MUST implement Book II's protected-deletion and tombstone model: they restrict accessibility or derived associations while preserving historical experience, immutable transition evidence, and Experience Ledger custody.

CMP-10 — Learning and Wisdom System

All CMP-10 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-11 — Temporal Service

All CMP-11 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-12 — Presence and Embodiment Layer

All CMP-12 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-13 — Capability Registry

All CMP-13 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-14 — Action and Transaction Engine

All CMP-14 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-15 — Verification Service

All CMP-15 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-16 — Simulation and Digital Twin Platform

All CMP-16 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-17 — Trust Service

All CMP-17 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-18 — Evidence Service

All CMP-18 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-19 — Privacy and Data Governance Service

All CMP-19 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-20 — Constitutional Firewall

All CMP-20 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-21 — Treaty Manager

All CMP-21 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-22 — Event and Messaging Platform

All CMP-22 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-23 — State and Persistence Platform

All CMP-23 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-24 — Observability and Audit Platform

All CMP-24 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-25 — Runtime Supervisor

All CMP-25 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-26 — Recovery Coordinator

All CMP-26 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-27 — Human Interaction Layer

All CMP-27 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-28 — Self-Model and Constitutional Mirror

All CMP-28 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

CMP-29 — Outcome and Success Evaluation Service

All CMP-29 contracts require `HAL-Internal-Authority-v1`, the HAL envelope, structured errors, semantic versioning, bounded limits, and conformance evidence. Their payload semantics remain exactly those stated by Book IV.

Appendix A — Error Code Registry

Appendix B — Contract Artifact Map

Appendix C — Certification Statement

Book IX v1.0 is certified as a contract-level implementation of Books I-IV, aligned with Book X. It introduces no component, state owner, constitutional principle, capability class, Treaty class, or Owner-authority change. Each Book IV logical-interface identifier maps to one and only one Book IX contract identifier. No Owner Review item is required.

Appendix D — Compact Contract Register

CMP-01 — Constitutional Kernel

Contract	Source	Kind	Operation / binding
IX-C-0001	CMP-01-IF-01	Command	Evaluate Protected Action — /hal/v1/cmp-01/evaluate-protected-action
IX-C-0002	CMP-01-IF-02	Command	Commit Protected Mutation — /hal/v1/cmp-01/commit-protected-mutation
IX-C-0003	CMP-01-IF-03	Command	Validate Owner Authorization — /hal/v1/cmp-01/validate-owner-authorization
IX-C-0004	CMP-01-IF-04	Command	Admit Constitutional Recovery — /hal/v1/cmp-01/admit-constitutional-recovery
IX-C-0005	CMP-01-IF-05	Query	Get Constitutional State — /hal/v1/cmp-01/get-constitutional-state
IX-C-0006	CMP-01-IF-06	Query	Get Protected Decision — /hal/v1/cmp-01/get-protected-decision
IX-C-0007	CMP-01-IF-07	Event	Protected Action Admitted — hal.cmp-01.protected.action.admitted.v1
IX-C-0008	CMP-01-IF-08	Event	Protected Action Denied — hal.cmp-01.protected.action.denied.v1
IX-C-0009	CMP-01-IF-09	Event	Constitutional Mode Changed — hal.cmp-01.constitutional.mode.changed.v1

CMP-02 — Identity and Continuity Service

Contract	Source	Kind	Operation / binding
IX-C-0010	CMP-02-IF-01	Command	Create Identity — /hal/v1/cmp-02/create-identity
IX-C-0011	CMP-02-IF-02	Command	Bind Identifier — /hal/v1/cmp-02/bind-identifier
IX-C-0012	CMP-02-IF-03	Command	Suspend Identity — /hal/v1/cmp-02/suspend-identity
IX-C-0013	CMP-02-IF-04	Command	Reconcile Continuity — /hal/v1/cmp-02/reconcile-continuity
IX-C-0014	CMP-02-IF-05	Query	Resolve Identity — /hal/v1/cmp-02/resolve-identity
IX-C-0015	CMP-02-IF-06	Query	Get Continuity Lineage — /hal/v1/cmp-02/get-continuity-lineage
IX-C-0016	CMP-02-IF-07	Query	Get Authentication Context — /hal/v1/cmp-02/get-authentication-context
IX-C-0017	CMP-02-IF-08	Event	Identity Created —

IX-C-0018	CMP-02-IF-09	Event	hal.cmp-02.identity.created.v1 Identity Suspended — hal.cmp-02.identity.suspended.v1
IX-C-0019	CMP-02-IF-10	Event	Continuity Reconciled — hal.cmp-02.continuity.reconciled.v1

CMP-03 — Authority and Delegation Service

Contract	Source	Kind	Operation / binding
IX-C-0020	CMP-03-IF-01	Command	Grant Delegation — /hal/v1/cmp-03/grant-delegation
IX-C-0021	CMP-03-IF-02	Command	Revoke Delegation — /hal/v1/cmp-03/revoke-delegation
IX-C-0022	CMP-03-IF-03	Command	Evaluate Permission — /hal/v1/cmp-03/evaluate-permission
IX-C-0023	CMP-03-IF-04	Command	Expire Delegation — /hal/v1/cmp-03/expire-delegation
IX-C-0024	CMP-03-IF-05	Query	Get Effective Authority — /hal/v1/cmp-03/get-effective-authority
IX-C-0025	CMP-03-IF-06	Query	Explain Policy Decision — /hal/v1/cmp-03/explain-policy-decision
IX-C-0026	CMP-03-IF-07	Query	List Active Delegations — /hal/v1/cmp-03/list-active-delegations
IX-C-0027	CMP-03-IF-08	Event	Delegation Activated — hal.cmp-03.delegation.activated.v1
IX-C-0028	CMP-03-IF-09	Event	Delegation Revoked — hal.cmp-03.delegation.revoked.v1
IX-C-0029	CMP-03-IF-10	Event	Permission Decided — hal.cmp-03.permission.decided.v1

CMP-04 — Intent Service

Contract	Source	Kind	Operation / binding
IX-C-0030	CMP-04-IF-01	Command	Register Intent — /hal/v1/cmp-04/register-intent
IX-C-0031	CMP-04-IF-02	Command	Revise Intent — /hal/v1/cmp-04/revise-intent
IX-C-0032	CMP-04-IF-03	Command	Decompose Goal — /hal/v1/cmp-04/decompose-goal
IX-C-0033	CMP-04-IF-04	Command	Retire Intent — /hal/v1/cmp-04/retire-intent
IX-C-0034	CMP-04-IF-05	Query	Get Intent Graph — /hal/v1/cmp-04/get-intent-graph
IX-C-0035	CMP-04-IF-06	Query	Trace Objective — /hal/v1/cmp-04/trace-objective
IX-C-0036	CMP-04-IF-07	Query	List Outcome Criteria — /hal/v1/cmp-04/list-outcome-criteria
IX-C-0037	CMP-04-IF-08	Event	Intent Registered — hal.cmp-04.intent.registered.v1

IX-C-0038	CMP-04-IF-09	Event	Intent Revised — hal.cmp-04.intent.revised.v1
IX-C-0039	CMP-04-IF-10	Event	Objective Retired — hal.cmp-04.objective.retired.v1

CMP-05 — Cognitive Orchestrator

Contract	Source	Kind	Operation / binding
IX-C-0040	CMP-05-IF-01	Command	Start Orchestration — /hal/v1/cmp-05/start-orchestration
IX-C-0041	CMP-05-IF-02	Command	Advance Execution Graph — /hal/v1/cmp-05/advance-execution-graph
IX-C-0042	CMP-05-IF-03	Command	Cancel Run — /hal/v1/cmp-05/cancel-run
IX-C-0043	CMP-05-IF-04	Command	Request Provider Work — /hal/v1/cmp-05/request-provider-work
IX-C-0044	CMP-05-IF-05	Query	Get Run State — /hal/v1/cmp-05/get-run-state
IX-C-0045	CMP-05-IF-06	Query	Explain Routing — /hal/v1/cmp-05/explain-routing
IX-C-0046	CMP-05-IF-07	Query	Get Execution Graph — /hal/v1/cmp-05/get-execution-graph
IX-C-0047	CMP-05-IF-08	Event	Run Started — hal.cmp-05.run.started.v1
IX-C-0048	CMP-05-IF-09	Event	Node Completed — hal.cmp-05.node.completed.v1
IX-C-0049	CMP-05-IF-10	Event	Run Failed — hal.cmp-05.run.failed.v1
IX-C-0050	CMP-05-IF-11	Event	Result Assembled — hal.cmp-05.result.assembled.v1

CMP-06 — Attention Manager

Contract	Source	Kind	Operation / binding
IX-C-0051	CMP-06-IF-01	Command	Submit Attention Request — /hal/v1/cmp-06/submit-attention-request
IX-C-0052	CMP-06-IF-02	Command	Admit Work — /hal/v1/cmp-06/admit-work
IX-C-0053	CMP-06-IF-03	Command	Preempt Work — /hal/v1/cmp-06/preempt-work
IX-C-0054	CMP-06-IF-04	Command	Release Placement — /hal/v1/cmp-06/release-placement
IX-C-0055	CMP-06-IF-05	Query	Get Queue State — /hal/v1/cmp-06/get-queue-state
IX-C-0056	CMP-06-IF-06	Query	Explain Priority — /hal/v1/cmp-06/explain-priority
IX-C-0057	CMP-06-IF-07	Query	Get Placement — /hal/v1/cmp-06/get-placement
IX-C-0058	CMP-06-IF-08	Event	Work Admitted — hal.cmp-06.work.admitted.v1
IX-C-0059	CMP-06-IF-09	Event	Work Preempted —

IX-C-0060	CMP-06-IF-10	Event	hal.cmp-06.work.preempted.v1 Placement Released — hal.cmp-06.placement.released.v1
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CMP-07 — Judgment Service

Contract	Source	Kind	Operation / binding
IX-C-0061	CMP-07-IF-01	Command	Request Judgment — /hal/v1/cmp-07/request-judgment
IX-C-0062	CMP-07-IF-02	Command	Record Decision — /hal/v1/cmp-07/record-decision
IX-C-0063	CMP-07-IF-03	Command	Reconsider Decision — /hal/v1/cmp-07/reconsider-decision
IX-C-0064	CMP-07-IF-04	Command	Withdraw Decision — /hal/v1/cmp-07/withdraw-decision
IX-C-0065	CMP-07-IF-05	Query	Get Decision Object — /hal/v1/cmp-07/get-decision-object
IX-C-0066	CMP-07-IF-06	Query	Explain Rationale — /hal/v1/cmp-07/explain-rationale
IX-C-0067	CMP-07-IF-07	Query	List Alternatives — /hal/v1/cmp-07/list-alternatives
IX-C-0068	CMP-07-IF-08	Event	Decision Recorded — hal.cmp-07.decision.recorded.v1
IX-C-0069	CMP-07-IF-09	Event	Decision Reconsidered — hal.cmp-07.decision.reconsidered.v1
IX-C-0070	CMP-07-IF-10	Event	Decision Withdrawn — hal.cmp-07.decision.withdrawn.v1

CMP-08 — Knowledge Service

Contract	Source	Kind	Operation / binding
IX-C-0071	CMP-08-IF-01	Command	Admit Knowledge Assertion — /hal/v1/cmp-08/admit-knowledge-assertion
IX-C-0072	CMP-08-IF-02	Command	Revise Knowledge Link — /hal/v1/cmp-08/revise-knowledge-link
IX-C-0073	CMP-08-IF-03	Command	Invalidate Assertion — /hal/v1/cmp-08/invalidate-assertion
IX-C-0074	CMP-08-IF-04	Command	Rebuild Index — /hal/v1/cmp-08/rebuild-index
IX-C-0075	CMP-08-IF-05	Query	Query Knowledge — /hal/v1/cmp-08/query-knowledge
IX-C-0076	CMP-08-IF-06	Query	Trace Provenance — /hal/v1/cmp-08/trace-provenance
IX-C-0077	CMP-08-IF-07	Query	Get Validity Context — /hal/v1/cmp-08/get-validity-context
IX-C-0078	CMP-08-IF-08	Event	Knowledge Admitted — hal.cmp-08.knowledge.admitted.v1
IX-C-0079	CMP-08-IF-09	Event	Assertion Invalidated —

IX-C-0080	CMP-08-IF-10	Event	hal.cmp-08.assertion.invalidated.v1 Index Rebuilt — hal.cmp-08.index.rebuilt.v1
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CMP-09 — Memory System

Contract	Source	Kind	Operation / binding
IX-C-0081	CMP-09-IF-01	Command	Store Memory Association — /hal/v1/cmp-09/store-memory-association
IX-C-0082	CMP-09-IF-02	Command	Revise Association — /hal/v1/cmp-09/revise-association
IX-C-0083	CMP-09-IF-03	Command	Restrict Memory Association — /hal/v1/cmp-09/restrict-memory-association
IX-C-0084	CMP-09-IF-04	Command	Reconcile Conversation — /hal/v1/cmp-09/reconcile-conversation
IX-C-0085	CMP-09-IF-05	Query	Recall Memory — /hal/v1/cmp-09/recall-memory
IX-C-0086	CMP-09-IF-06	Query	Trace Memory Source — /hal/v1/cmp-09/trace-memory-source
IX-C-0087	CMP-09-IF-07	Query	Get Retention Status — /hal/v1/cmp-09/get-retention-status
IX-C-0088	CMP-09-IF-08	Event	Memory Associated — hal.cmp-09.memory.associated.v1
IX-C-0089	CMP-09-IF-09	Event	Memory Association Restricted — hal.cmp-09.memory.association-restricted.v1
IX-C-0090	CMP-09-IF-10	Event	Conversation Reconciled — hal.cmp-09.conversation.reconciled.v1

CMP-10 — Learning and Wisdom System

Contract	Source	Kind	Operation / binding
IX-C-0091	CMP-10-IF-01	Command	Record Experience — /hal/v1/cmp-10/record-experience
IX-C-0092	CMP-10-IF-02	Command	Propose Pattern — /hal/v1/cmp-10/propose-pattern
IX-C-0093	CMP-10-IF-03	Command	Promote Pattern — /hal/v1/cmp-10/promote-pattern
IX-C-0094	CMP-10-IF-04	Command	Retire Wisdom — /hal/v1/cmp-10/retire-wisdom
IX-C-0095	CMP-10-IF-05	Query	Get Experience — /hal/v1/cmp-10/get-experience
IX-C-0096	CMP-10-IF-06	Query	Explain Pattern — /hal/v1/cmp-10/explain-pattern
IX-C-0097	CMP-10-IF-07	Query	Get Wisdom Limits — /hal/v1/cmp-10/get-wisdom-limits
IX-C-0098	CMP-10-IF-08	Event	Experience Recorded — hal.cmp-10.experience.recorded.v1

IX-C-0099	CMP-10-IF-09	Event	Pattern Promoted — hal.cmp-10.pattern.promoted.v1
IX-C-0100	CMP-10-IF-10	Event	Wisdom Retired — hal.cmp-10.wisdom.retired.v1

CMP-11 — Temporal Service

Contract	Source	Kind	Operation / binding
IX-C-0101	CMP-11-IF-01	Command	Issue Lease — /hal/v1/cmp-11/issue-lease
IX-C-0102	CMP-11-IF-02	Command	Renew Lease — /hal/v1/cmp-11/renew-lease
IX-C-0103	CMP-11-IF-03	Command	Record Logical Time — /hal/v1/cmp-11/record-logical-time
IX-C-0104	CMP-11-IF-04	Command	Declare Time Source Degraded — /hal/v1/cmp-11/declare-time-source-degraded
IX-C-0105	CMP-11-IF-05	Query	Get Current Time Context — /hal/v1/cmp-11/get-current-time-context
IX-C-0106	CMP-11-IF-06	Query	Compare Causality — /hal/v1/cmp-11/compare-causality
IX-C-0107	CMP-11-IF-07	Query	Get Freshness — /hal/v1/cmp-11/get-freshness
IX-C-0108	CMP-11-IF-08	Event	Lease Issued — hal.cmp-11.lease.issued.v1
IX-C-0109	CMP-11-IF-09	Event	Lease Expired — hal.cmp-11.lease.expired.v1
IX-C-0110	CMP-11-IF-10	Event	Time Source Degraded — hal.cmp-11.time.source.degraded.v1

CMP-12 — Presence and Embodiment Layer

Contract	Source	Kind	Operation / binding
IX-C-0111	CMP-12-IF-01	Command	Create Presence — /hal/v1/cmp-12/create-presence
IX-C-0112	CMP-12-IF-02	Command	Bind Embodiment — /hal/v1/cmp-12/bind-embodiment
IX-C-0113	CMP-12-IF-03	Command	Handoff Presence — /hal/v1/cmp-12/handoff-presence
IX-C-0114	CMP-12-IF-04	Command	End Presence — /hal/v1/cmp-12/end-presence
IX-C-0115	CMP-12-IF-05	Query	Get Presence Context — /hal/v1/cmp-12/get-presence-context
IX-C-0116	CMP-12-IF-06	Query	List Embodiments — /hal/v1/cmp-12/list-embodiments
IX-C-0117	CMP-12-IF-07	Query	Get Audience Context — /hal/v1/cmp-12/get-audience-context
IX-C-0118	CMP-12-IF-08	Event	Presence Created — hal.cmp-12.presence.created.v1
IX-C-0119	CMP-12-IF-09	Event	Embodiment Bound —

IX-C-0120	CMP-12-IF-10	Event	hal.cmp-12.embodiment.bound.v1 Presence Handed Off — hal.cmp-12.presence.handed.off.v1
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CMP-13 — Capability Registry

Contract	Source	Kind	Operation / binding
IX-C-0121	CMP-13-IF-01	Command	Register Capability — /hal/v1/cmp-13/register-capability
IX-C-0122	CMP-13-IF-02	Command	Register Provider — /hal/v1/cmp-13/register-provider
IX-C-0123	CMP-13-IF-03	Command	Qualify Provider — /hal/v1/cmp-13/qualify-provider
IX-C-0124	CMP-13-IF-04	Command	Retire Adapter — /hal/v1/cmp-13/retire-adapter
IX-C-0125	CMP-13-IF-05	Query	Discover Capability — /hal/v1/cmp-13/discover-capability
IX-C-0126	CMP-13-IF-06	Query	Get Provider Qualification — /hal/v1/cmp-13/get-provider-qualification
IX-C-0127	CMP-13-IF-07	Query	Resolve Adapter — /hal/v1/cmp-13/resolve-adapter
IX-C-0128	CMP-13-IF-08	Event	Capability Registered — hal.cmp-13.capability.registered.v1
IX-C-0129	CMP-13-IF-09	Event	Provider Qualified — hal.cmp-13.provider.qualified.v1
IX-C-0130	CMP-13-IF-10	Event	Provider Quarantined — hal.cmp-13.provider.quarantined.v1

CMP-14 — Action and Transaction Engine

Contract	Source	Kind	Operation / binding
IX-C-0131	CMP-14-IF-01	Command	Propose Action — /hal/v1/cmp-14/propose-action
IX-C-0132	CMP-14-IF-02	Command	Prepare Transaction — /hal/v1/cmp-14/prepare-transaction
IX-C-0133	CMP-14-IF-03	Command	Commit Transaction — /hal/v1/cmp-14/commit-transaction
IX-C-0134	CMP-14-IF-04	Command	Rollback Transaction — /hal/v1/cmp-14/rollback-transaction
IX-C-0135	CMP-14-IF-05	Command	Begin Compensation — /hal/v1/cmp-14/begin-compensation
IX-C-0136	CMP-14-IF-06	Query	Get Transaction State — /hal/v1/cmp-14/get-transaction-state
IX-C-0137	CMP-14-IF-07	Query	Explain Commit Decision — /hal/v1/cmp-14/explain-commit-decision
IX-C-0138	CMP-14-IF-08	Query	Get Idempotency Result —

IX-C-0139	CMP-14-IF-09	Event	/hal/v1/cmp-14/get-idempotency-result Action Authorized — hal.cmp-14.action.authorized.v1 Transaction Committed — hal.cmp-14.transaction.committed.v1 Transaction Rolled Back — hal.cmp-14.transaction.rolled.back.v1 Compensation Started — hal.cmp-14.compensation.started.v1
IX-C-0140	CMP-14-IF-10	Event	
IX-C-0141	CMP-14-IF-11	Event	
IX-C-0142	CMP-14-IF-12	Event	

CMP-15 — Verification Service

Contract	Source	Kind	Operation / binding
IX-C-0143	CMP-15-IF-01	Command	Start Verification — /hal/v1/cmp-15/start-verification
IX-C-0144	CMP-15-IF-02	Command	Record Verification Result — /hal/v1/cmp-15/record-verification-result
IX-C-0145	CMP-15-IF-03	Command	Invalidate Result — /hal/v1/cmp-15/invalidate-result
IX-C-0146	CMP-15-IF-04	Command	Request Reproduction — /hal/v1/cmp-15/request-reproduction
IX-C-0147	CMP-15-IF-05	Query	Get Verification Result — /hal/v1/cmp-15/get-verification-result
IX-C-0148	CMP-15-IF-06	Query	Explain Confidence — /hal/v1/cmp-15/explain-confidence
IX-C-0149	CMP-15-IF-07	Query	List Defeaters — /hal/v1/cmp-15/list-defeaters
IX-C-0150	CMP-15-IF-08	Event	Verification Started — hal.cmp-15.verification.started.v1
IX-C-0151	CMP-15-IF-09	Event	Claim Verified — hal.cmp-15.claim.verified.v1
IX-C-0152	CMP-15-IF-10	Event	Claim Falsified — hal.cmp-15.claim.falsified.v1
IX-C-0153	CMP-15-IF-11	Event	Result Invalidated — hal.cmp-15.result.invalidated.v1

CMP-16 — Simulation and Digital Twin Platform

Contract	Source	Kind	Operation / binding
IX-C-0154	CMP-16-IF-01	Command	Create Simulation — /hal/v1/cmp-16/create-simulation
IX-C-0155	CMP-16-IF-02	Command	Load Twin State — /hal/v1/cmp-16/load-twin-state
IX-C-0156	CMP-16-IF-03	Command	Run Scenario — /hal/v1/cmp-16/run-scenario
IX-C-0157	CMP-16-IF-04	Command	Inject Failure — /hal/v1/cmp-16/inject-failure
IX-C-0158	CMP-16-IF-05	Command	Terminate Simulation — /hal/v1/cmp-16/terminate-

IX-C-0159	CMP-16-IF-06	Query	simulation Get Scenario Result — /hal/v1/cmp-16/get-scenario- result
IX-C-0160	CMP-16-IF-07	Query	Get Fidelity Score — /hal/v1/cmp-16/get-fidelity-score
IX-C-0161	CMP-16-IF-08	Query	Get Isolation Status — /hal/v1/cmp-16/get-isolation- status
IX-C-0162	CMP-16-IF-09	Event	Simulation Started — hal.cmp-16.simulation.started.v1
IX-C-0163	CMP-16-IF-10	Event	Scenario Completed — hal.cmp-16.scenario.completed.v1
IX-C-0164	CMP-16-IF-11	Event	Isolation Breach Detected — hal.cmp-16.isolation.breach.detec- ted.v1

CMP-17 — Trust Service

Contract	Source	Kind	Operation / binding
IX-C-0165	CMP-17-IF-01	Command	Request Trust Assessment — /hal/v1/cmp-17/request-trust- assessment
IX-C-0166	CMP-17-IF-02	Command	Update Trust Dimension — /hal/v1/cmp-17/update-trust- dimension
IX-C-0167	CMP-17-IF-03	Command	Invalidate Assessment — /hal/v1/cmp-17/invalidate- assessment
IX-C-0168	CMP-17-IF-04	Query	Get Trust Assessment — /hal/v1/cmp-17/get-trust- assessment
IX-C-0169	CMP-17-IF-05	Query	Explain Trust Basis — /hal/v1/cmp-17/explain-trust- basis
IX-C-0170	CMP-17-IF-06	Query	List Expiring Assessments — /hal/v1/cmp-17/list-expiring- assessments
IX-C-0171	CMP-17-IF-07	Event	Trust Assessed — hal.cmp-17.trust.assessed.v1
IX-C-0172	CMP-17-IF-08	Event	Trust Degraded — hal.cmp-17.trust.degraded.v1
IX-C-0173	CMP-17-IF-09	Event	Assessment Expired — hal.cmp-17.assessment.expired.v1

CMP-18 — Evidence Service

Contract	Source	Kind	Operation / binding
IX-C-0174	CMP-18-IF-01	Command	Submit Evidence Candidate — /hal/v1/cmp-18/submit-evidence- candidate
IX-C-0175	CMP-18-IF-02	Command	Admit Evidence Object — /hal/v1/cmp-18/admit-evidence- object
IX-C-0176	CMP-18-IF-03	Command	Challenge Evidence — /hal/v1/cmp-18/challenge- evidence

IX-C-0177	CMP-18-IF-04	Command	Record Supersession — /hal/v1/cmp-18/record-supersession
IX-C-0178	CMP-18-IF-05	Query	Get Evidence Object — /hal/v1/cmp-18/get-evidence-object
IX-C-0179	CMP-18-IF-06	Query	Trace Custody — /hal/v1/cmp-18/trace-custody
IX-C-0180	CMP-18-IF-07	Query	Verify Evidence Integrity — /hal/v1/cmp-18/verify-evidence-integrity
IX-C-0181	CMP-18-IF-08	Event	Evidence Admitted — hal.cmp-18.evidence.admitted.v1
IX-C-0182	CMP-18-IF-09	Event	Evidence Challenged — hal.cmp-18.evidence.challenged.v1
IX-C-0183	CMP-18-IF-10	Event	Evidence Superseded — hal.cmp-18.evidence.superseded.v1

CMP-19 — Privacy and Data Governance Service

Contract	Source	Kind	Operation / binding
IX-C-0184	CMP-19-IF-01	Command	Classify Data — /hal/v1/cmp-19/classify-data
IX-C-0185	CMP-19-IF-02	Command	Authorize Purpose — /hal/v1/cmp-19/authorize-purpose
IX-C-0186	CMP-19-IF-03	Command	Set Retention — /hal/v1/cmp-19/set-retention
IX-C-0187	CMP-19-IF-04	Command	Order Deletion — /hal/v1/cmp-19/order-deletion
IX-C-0188	CMP-19-IF-05	Command	Restrict Disclosure — /hal/v1/cmp-19/restrict-disclosure
IX-C-0189	CMP-19-IF-06	Query	Get Data Policy — /hal/v1/cmp-19/get-data-policy
IX-C-0190	CMP-19-IF-07	Query	Evaluate Data Use — /hal/v1/cmp-19/evaluate-data-use
IX-C-0191	CMP-19-IF-08	Query	Get Deletion Status — /hal/v1/cmp-19/get-deletion-status
IX-C-0192	CMP-19-IF-09	Event	Data Classified — hal.cmp-19.data.classified.v1
IX-C-0193	CMP-19-IF-10	Event	Purpose Authorized — hal.cmp-19.purpose.authorized.v1
IX-C-0194	CMP-19-IF-11	Event	Deletion Completed — hal.cmp-19.deletion.completed.v1
IX-C-0195	CMP-19-IF-12	Event	Use Denied — hal.cmp-19.use.denied.v1

CMP-20 — Constitutional Firewall

Contract	Source	Kind	Operation / binding
IX-C-0196	CMP-20-IF-01	Command	Open Domain Session — /hal/v1/cmp-20/open-domain-session
IX-C-0197	CMP-20-IF-02	Command	Authorize Exchange —

IX-C-0198	CMP-20-IF-03	Command	/hal/v1/cmp-20/authorize-exchange Terminate Exchange — /hal/v1/cmp-20/terminate-exchange
IX-C-0199	CMP-20-IF-04	Command	Quarantine Payload — /hal/v1/cmp-20/quarantine-payload
IX-C-0200	CMP-20-IF-05	Query	Get Firewall Decision — /hal/v1/cmp-20/get-firewall-decision
IX-C-0201	CMP-20-IF-06	Query	Get Active Treaty View — /hal/v1/cmp-20/get-active-treaty-view
IX-C-0202	CMP-20-IF-07	Query	Explain Denial — /hal/v1/cmp-20/explain-denial
IX-C-0203	CMP-20-IF-08	Event	Exchange Allowed — hal.cmp-20.exchange.allowed.v1
IX-C-0204	CMP-20-IF-09	Event	Exchange Denied — hal.cmp-20.exchange.denied.v1
IX-C-0205	CMP-20-IF-10	Event	Domain Session Terminated — hal.cmp-20.domain.session.terminated.v1

CMP-21 — Treaty Manager

Contract	Source	Kind	Operation / binding
IX-C-0206	CMP-21-IF-01	Command	Propose Treaty — /hal/v1/cmp-21/propose-treaty
IX-C-0207	CMP-21-IF-02	Command	Authorize Treaty — /hal/v1/cmp-21/authorize-treaty
IX-C-0208	CMP-21-IF-03	Command	Activate Treaty — /hal/v1/cmp-21/activate-treaty
IX-C-0209	CMP-21-IF-04	Command	Suspend Treaty — /hal/v1/cmp-21/suspend-treaty
IX-C-0210	CMP-21-IF-05	Command	Revoke Treaty — /hal/v1/cmp-21/revoke-treaty
IX-C-0211	CMP-21-IF-06	Query	Get Treaty — /hal/v1/cmp-21/get-treaty
IX-C-0212	CMP-21-IF-07	Query	Compare Treaty Version — /hal/v1/cmp-21/compare-treaty-version
IX-C-0213	CMP-21-IF-08	Query	List Active Treaties — /hal/v1/cmp-21/list-active-treaties
IX-C-0214	CMP-21-IF-09	Event	Treaty Proposed — hal.cmp-21.treaty.proposed.v1
IX-C-0215	CMP-21-IF-10	Event	Treaty Activated — hal.cmp-21.treaty.activated.v1
IX-C-0216	CMP-21-IF-11	Event	Treaty Suspended — hal.cmp-21.treaty.suspended.v1
IX-C-0217	CMP-21-IF-12	Event	Treaty Revoked — hal.cmp-21.treaty.revoked.v1

CMP-22 — Event and Messaging Platform

Contract	Source	Kind	Operation / binding
IX-C-0218	CMP-22-IF-01	Command	Publish Command —

IX-C-0219	CMP-22-IF-02	Command	/hal/v1/cmp-22/publish-command Append Event — /hal/v1/cmp-22/append-event
IX-C-0220	CMP-22-IF-03	Command	Acknowledge Delivery — /hal/v1/cmp-22/acknowledge-delivery
IX-C-0221	CMP-22-IF-04	Command	Quarantine Message — /hal/v1/cmp-22/quarantine-message
IX-C-0222	CMP-22-IF-05	Query	Query Event Stream — /hal/v1/cmp-22/query-event-stream
IX-C-0223	CMP-22-IF-06	Query	Get Delivery State — /hal/v1/cmp-22/get-delivery-state
IX-C-0224	CMP-22-IF-07	Query	Trace Causation — /hal/v1/cmp-22/trace-causation
IX-C-0225	CMP-22-IF-08	Event	Message Accepted — hal.cmp-22.message.accepted.v1
IX-C-0226	CMP-22-IF-09	Event	Event Published — hal.cmp-22.event.published.v1
IX-C-0227	CMP-22-IF-10	Event	Delivery Failed — hal.cmp-22.delivery.failed.v1
IX-C-0228	CMP-22-IF-11	Event	Message Quarantined — hal.cmp-22.message.quarantined.v1

CMP-23 — State and Persistence Platform

Contract	Source	Kind	Operation / binding
IX-C-0229	CMP-23-IF-01	Command	Commit State Change — /hal/v1/cmp-23/commit-state-change
IX-C-0230	CMP-23-IF-02	Command	Create Snapshot — /hal/v1/cmp-23/create-snapshot
IX-C-0231	CMP-23-IF-03	Command	Migrate Store — /hal/v1/cmp-23/migrate-store
IX-C-0232	CMP-23-IF-04	Command	Rebuild Projection — /hal/v1/cmp-23/rebuild-projection
IX-C-0233	CMP-23-IF-05	Query	Read State — /hal/v1/cmp-23/read-state
IX-C-0234	CMP-23-IF-06	Query	Get Version — /hal/v1/cmp-23/get-version
IX-C-0235	CMP-23-IF-07	Query	Verify Replica — /hal/v1/cmp-23/verify-replica
IX-C-0236	CMP-23-IF-08	Query	Get Migration State — /hal/v1/cmp-23/get-migration-state
IX-C-0237	CMP-23-IF-09	Event	State Committed — hal.cmp-23.state.committed.v1
IX-C-0238	CMP-23-IF-10	Event	Snapshot Created — hal.cmp-23.snapshot.created.v1
IX-C-0239	CMP-23-IF-11	Event	Migration Completed — hal.cmp-23.migration.completed.v1
IX-C-0240	CMP-23-IF-12	Event	Replica Diverged — hal.cmp-23.replica.diverged.v1

CMP-24 — Observability and Audit Platform

Contract	Source	Kind	Operation / binding
IX-C-0241	CMP-24-IF-01	Command	Append Audit Record — /hal/v1/cmp-24/append-audit-record
IX-C-0242	CMP-24-IF-02	Command	Emit Telemetry — /hal/v1/cmp-24/emit-telemetry
IX-C-0243	CMP-24-IF-03	Command	Declare Alert — /hal/v1/cmp-24/declare-alert
IX-C-0244	CMP-24-IF-04	Command	Seal Audit Segment — /hal/v1/cmp-24/seal-audit-segment
IX-C-0245	CMP-24-IF-05	Query	Query Audit Ledger — /hal/v1/cmp-24/query-audit-ledger
IX-C-0246	CMP-24-IF-06	Query	Get Trace — /hal/v1/cmp-24/get-trace
IX-C-0247	CMP-24-IF-07	Query	Get Metric — /hal/v1/cmp-24/get-metric
IX-C-0248	CMP-24-IF-08	Query	Verify Audit Chain — /hal/v1/cmp-24/verify-audit-chain
IX-C-0249	CMP-24-IF-09	Event	Audit Record Appended — hal.cmp-24.audit.record.appended.v1
IX-C-0250	CMP-24-IF-10	Event	Alert Raised — hal.cmp-24.alert.raised.v1
IX-C-0251	CMP-24-IF-11	Event	Audit Chain Broken — hal.cmp-24.audit.chain.broken.v1

CMP-25 — Runtime Supervisor

Contract	Source	Kind	Operation / binding
IX-C-0252	CMP-25-IF-01	Command	Start Service — /hal/v1/cmp-25/start-service
IX-C-0253	CMP-25-IF-02	Command	Stop Service — /hal/v1/cmp-25/stop-service
IX-C-0254	CMP-25-IF-03	Command	Restart Service — /hal/v1/cmp-25/restart-service
IX-C-0255	CMP-25-IF-04	Command	Enter Degraded Mode — /hal/v1/cmp-25/enter-degraded-mode
IX-C-0256	CMP-25-IF-05	Command	Quarantine Service — /hal/v1/cmp-25/quarantine-service
IX-C-0257	CMP-25-IF-06	Query	Get Service State — /hal/v1/cmp-25/get-service-state
IX-C-0258	CMP-25-IF-07	Query	Get Health — /hal/v1/cmp-25/get-health
IX-C-0259	CMP-25-IF-08	Query	Get Resource Use — /hal/v1/cmp-25/get-resource-use
IX-C-0260	CMP-25-IF-09	Event	Service Ready — hal.cmp-25.service.ready.v1
IX-C-0261	CMP-25-IF-10	Event	Service Degraded — hal.cmp-25.service.degraded.v1
IX-C-0262	CMP-25-IF-11	Event	Service Quarantined — hal.cmp-25.service.quarantined.v1
IX-C-0263	CMP-25-IF-12	Event	Service Stopped —

hal.cmp-25.service.stopped.v1

CMP-26 — Recovery Coordinator

Contract	Source	Kind	Operation / binding
IX-C-0264	CMP-26-IF-01	Command	Open Recovery Case — /hal/v1/cmp-26/open-recovery-case
IX-C-0265	CMP-26-IF-02	Command	Authorize Recovery Step — /hal/v1/cmp-26/authorize-recovery-step
IX-C-0266	CMP-26-IF-03	Command	Restore State — /hal/v1/cmp-26/restore-state
IX-C-0267	CMP-26-IF-04	Command	Reconcile Partition — /hal/v1/cmp-26/reconcile-partition
IX-C-0268	CMP-26-IF-05	Command	Close Recovery — /hal/v1/cmp-26/close-recovery
IX-C-0269	CMP-26-IF-06	Query	Get Recovery State — /hal/v1/cmp-26/get-recovery-state
IX-C-0270	CMP-26-IF-07	Query	Get RPO/RTO Result — /hal/v1/cmp-26/get-rpo-rto-result
IX-C-0271	CMP-26-IF-08	Query	Explain Reconciliation — /hal/v1/cmp-26/explain-reconciliation
IX-C-0272	CMP-26-IF-09	Event	Recovery Started — hal.cmp-26.recovery.started.v1
IX-C-0273	CMP-26-IF-10	Event	State Restored — hal.cmp-26.state.restored.v1
IX-C-0274	CMP-26-IF-11	Event	Partition Reconciled — hal.cmp-26.partition.reconciled.v1
IX-C-0275	CMP-26-IF-12	Event	Recovery Failed — hal.cmp-26.recovery.failed.v1

CMP-27 — Human Interaction Layer

Contract	Source	Kind	Operation / binding
IX-C-0276	CMP-27-IF-01	Command	Start Interaction — /hal/v1/cmp-27/start-interaction
IX-C-0277	CMP-27-IF-02	Command	Submit Human Input — /hal/v1/cmp-27/submit-human-input
IX-C-0278	CMP-27-IF-03	Command	Request Confirmation — /hal/v1/cmp-27/request-confirmation
IX-C-0279	CMP-27-IF-04	Command	End Interaction — /hal/v1/cmp-27/end-interaction
IX-C-0280	CMP-27-IF-05	Query	Get Session State — /hal/v1/cmp-27/get-session-state
IX-C-0281	CMP-27-IF-06	Query	Get Accessible Representation — /hal/v1/cmp-27/get-accessible-representation
IX-C-0282	CMP-27-IF-07	Query	Get Action Status — /hal/v1/cmp-27/get-action-status
IX-C-0283	CMP-27-IF-08	Event	Interaction Started — hal.cmp-27.interaction.started.v1

IX-C-0284	CMP-27-IF-09	Event	Clarification Requested — hal.cmp-27.clarification.requested.v1
IX-C-0285	CMP-27-IF-10	Event	Confirmation Recorded — hal.cmp-27.confirmation.recorded.v1
IX-C-0286	CMP-27-IF-11	Event	Interaction Ended — hal.cmp-27.interaction.ended.v1

CMP-28 — Self-Model and Constitutional Mirror

Contract	Source	Kind	Operation / binding
IX-C-0287	CMP-28-IF-01	Command	Refresh Self Model — /hal/v1/cmp-28/refresh-self-model
IX-C-0288	CMP-28-IF-02	Command	Publish Constitutional Mirror — /hal/v1/cmp-28/publish-constitutional-mirror
IX-C-0289	CMP-28-IF-03	Command	Invalidate Projection — /hal/v1/cmp-28/invalidate-projection
IX-C-0290	CMP-28-IF-04	Query	Get Self Description — /hal/v1/cmp-28/get-self-description
IX-C-0291	CMP-28-IF-05	Query	Trace Mirror Claim — /hal/v1/cmp-28/trace-mirror-claim
IX-C-0292	CMP-28-IF-06	Query	Get Declared Limitation — /hal/v1/cmp-28/get-declared-limitation
IX-C-0293	CMP-28-IF-07	Event	Self Model Refreshed — hal.cmp-28.self.model.refreshed.v1
IX-C-0294	CMP-28-IF-08	Event	Mirror Published — hal.cmp-28.mirror.published.v1
IX-C-0295	CMP-28-IF-09	Event	Projection Invalidated — hal.cmp-28.projection.invalidated.v1

CMP-29 — Outcome and Success Evaluation Service

Contract	Source	Kind	Operation / binding
IX-C-0296	CMP-29-IF-01	Command	Open Outcome Evaluation — /hal/v1/cmp-29/open-outcome-evaluation
IX-C-0297	CMP-29-IF-02	Command	Record Outcome — /hal/v1/cmp-29/record-outcome
IX-C-0298	CMP-29-IF-03	Command	Evaluate Success — /hal/v1/cmp-29/evaluate-success
IX-C-0299	CMP-29-IF-04	Command	Refer Learning — /hal/v1/cmp-29/refer-learning
IX-C-0300	CMP-29-IF-05	Query	Get Outcome Object — /hal/v1/cmp-29/get-outcome-object
IX-C-0301	CMP-29-IF-06	Query	Explain Success Result — /hal/v1/cmp-29/explain-success-result

IX-C-0302	CMP-29-IF-07	Query	List Side Effects — /hal/v1/cmp-29/list-side-effects
IX-C-0303	CMP-29-IF-08	Event	Outcome Recorded — hal.cmp-29.outcome.recorded.v1
IX-C-0304	CMP-29-IF-09	Event	Success Evaluated — hal.cmp-29.success.evaluated.v1
IX-C-0305	CMP-29-IF-10	Event	Review Required — hal.cmp-29.review.required.v1